Bangladesh Army International University of Science and Technology

Department of Computer Science and Engineering

Project Report

Project Name : **Taskly - Task Management System**

Course Title : Database Management System Sessional.

Course Code : CSE 314

Level : 3 **Term** : I **Section** : A **Group** : 1

Date of Submission : 12/1/2026 **Semester** : Fall **Year** : 2025

Submitted to:

Mofrad Jamil Joy
Lecturer
Dept. of CSE, BAIUST

Submitted by:

Name : **Nosratee Jahan Naba**
Student ID : **1118005**

Name : **Habiba Sultana Bristy**
Student ID : **1118003**

Name : **Mst. Saima Rahman Rabaya**
Student ID : **1118012**

Name : **Amina Islam Meem**
Student ID : **1118019**

1. Introduction

Problem Domain

In modern organizations, managing tasks, projects, and team collaboration efficiently is a major challenge. Tasks are often scattered across emails, spreadsheets, and chat applications, resulting in poor visibility and communication gaps. Managers struggle to track progress, responsibilities become unclear, and workflows become inefficient. These issues lead to missed deadlines, duplicated work, and reduced productivity. Therefore, a centralized task management system is essential.

Purpose and Objectives of the Project

The primary purpose of this project is to develop a **web-based Task Management System** that helps teams organize tasks, collaborate effectively, and monitor progress in real time.

Objectives:

- Provide centralized task creation, update, and deletion
- Implement role-based access control (Admin, Manager, Employee)
- Track task status and priority in real time
- Display task statistics through dashboards
- Ensure a user-friendly and responsive interface

Why a Database Solution is Needed

A database solution is required to ensure reliable and efficient system operation.

- **Data Persistence:** Stores tasks, projects, and user data permanently
- **Relationship Management:** Manages relationships among users, tasks, and projects
- **Data Integrity:** Ensures accuracy using constraints and validations
- **Concurrent Access:** Supports multiple users accessing data simultaneously

Using a database ensures security, scalability, consistency, and efficient data management, which cannot be reliably achieved using files or spreadsheets.

2. System overview

Description of the Proposed System

Taskly is a web-based task management system developed using the Django framework. The system helps organizations efficiently manage projects, tasks, and team collaboration through a centralized platform.

The application follows Django's **MVT (Model–View–Template)** architecture.

- The **Model layer** defines database entities such as User, Project, Task, and TaskDetail and manages relationships.
- The **View layer** handles user requests, authentication, authorization, and business logic.
- The **Template layer** presents data using HTML and Tailwind CSS with responsive design.

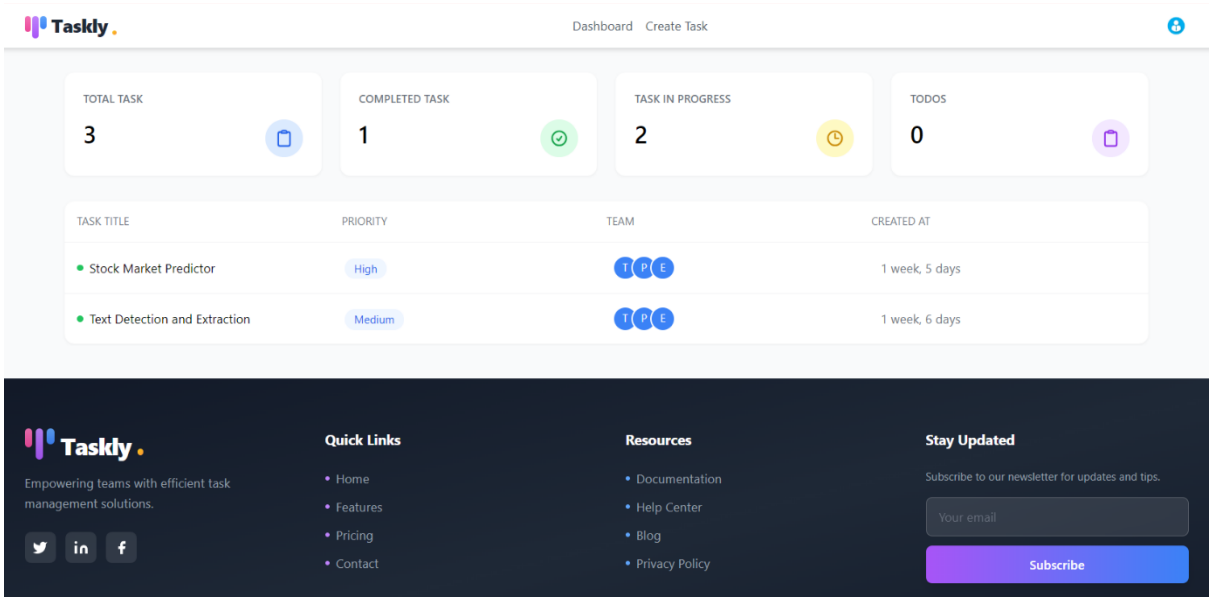
The system workflow starts with user authentication, followed by role-based access (Manager, Employee, Admin). Users can create, view, update, or delete tasks based on their permissions. Automated email notifications are sent when tasks are assigned.

Technology Stack:

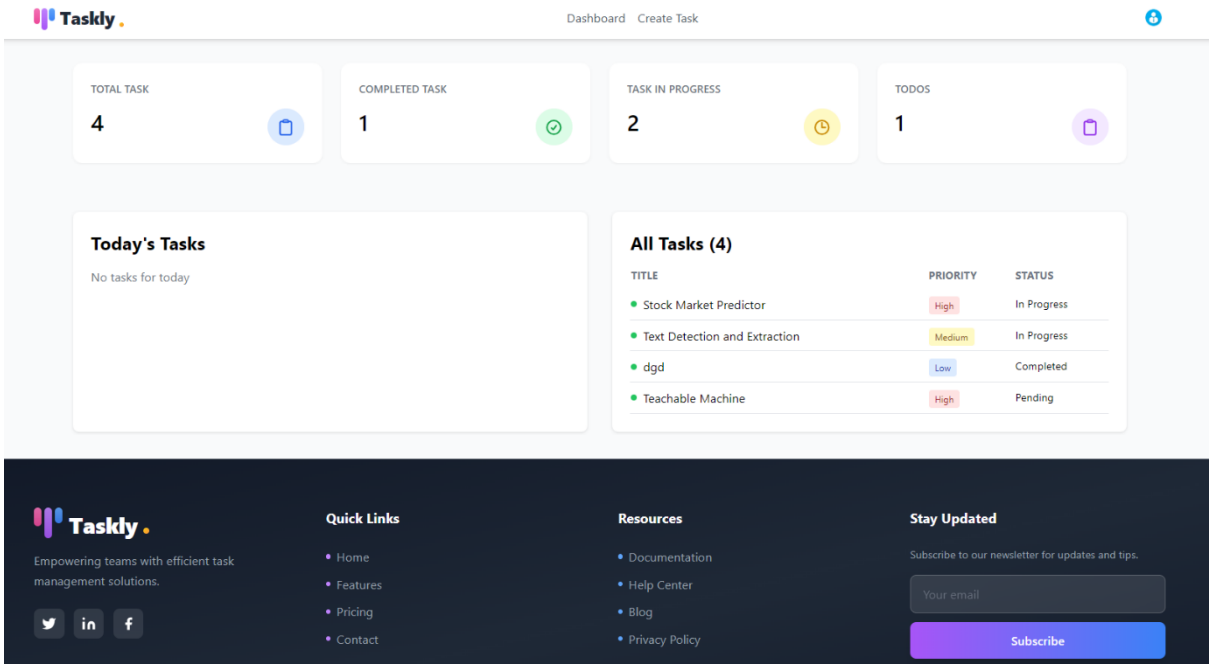
- Backend: Django
- Frontend: HTML5, Tailwind CSS
- Database: PostgreSQL
- Authentication: Django Auth

Key Features and Functionalities

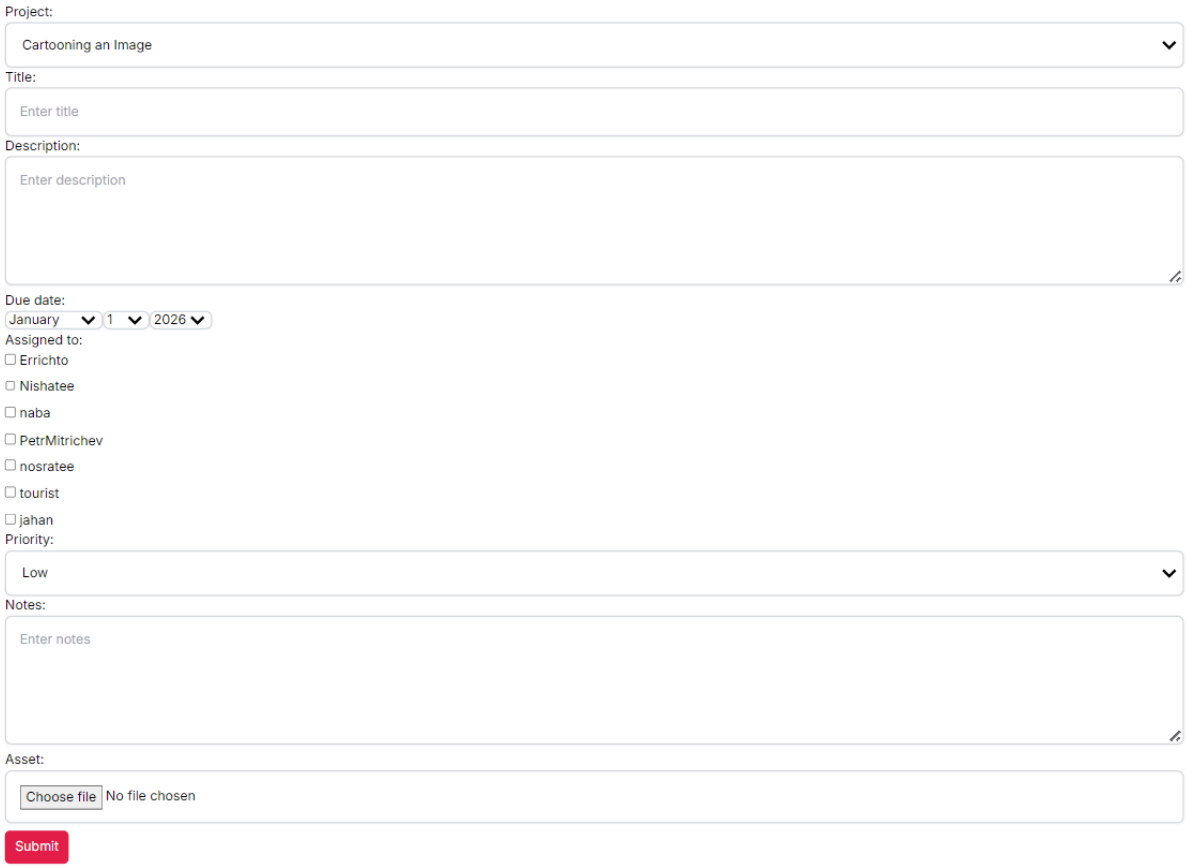
- **Filtering :** Tasks can be filtered by status (Pending, In Progress, Completed, Todos).
- **Team Assignment:** Supports assigning multiple employees to a task.
- **Manager Dashboard:** Displays task statistics and allows full task management



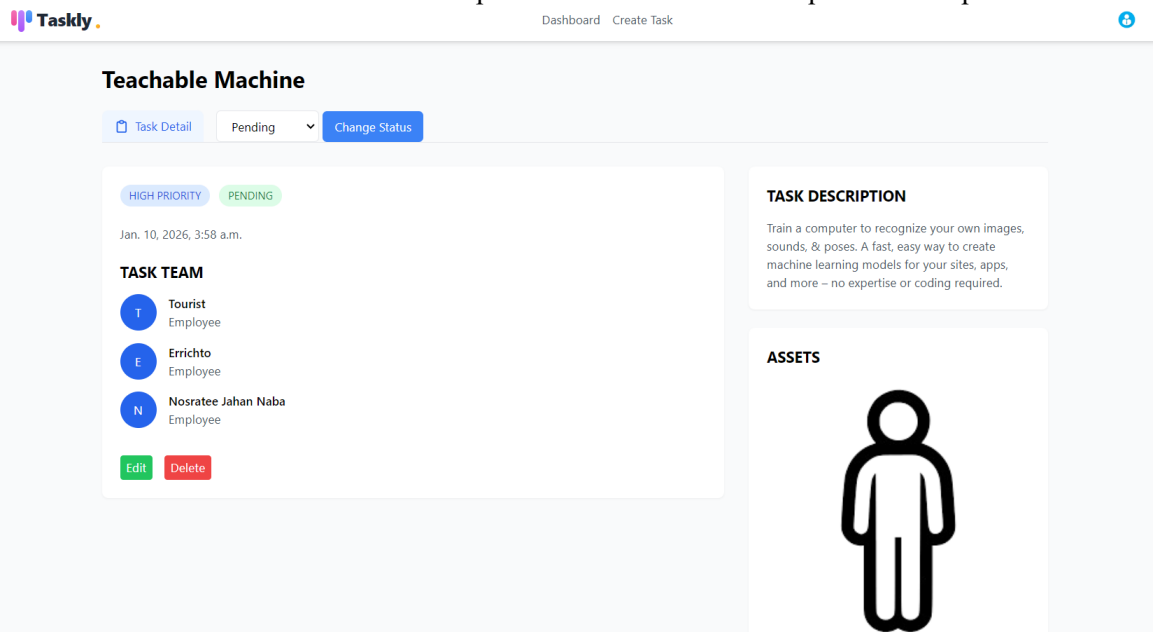
- **Employee Dashboard:** Displays task statistics.



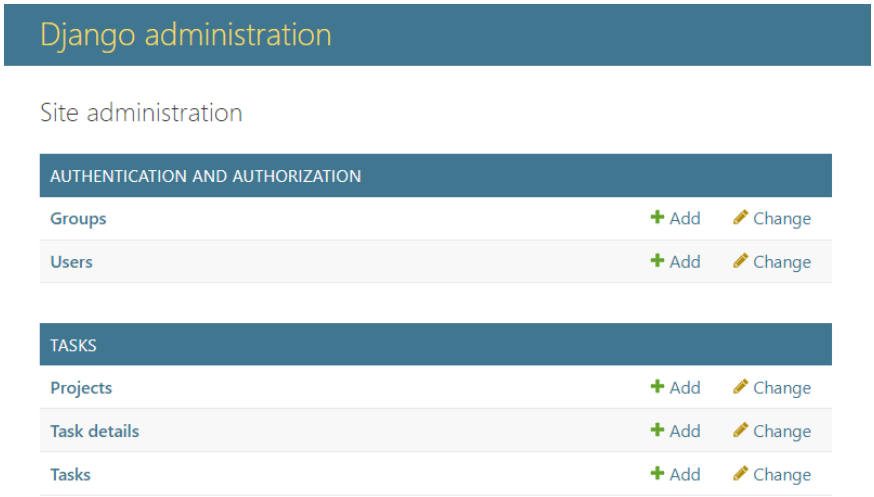
- **Task Creation & Update:** Forms to create and modify tasks with priority, due date, and team assignment.



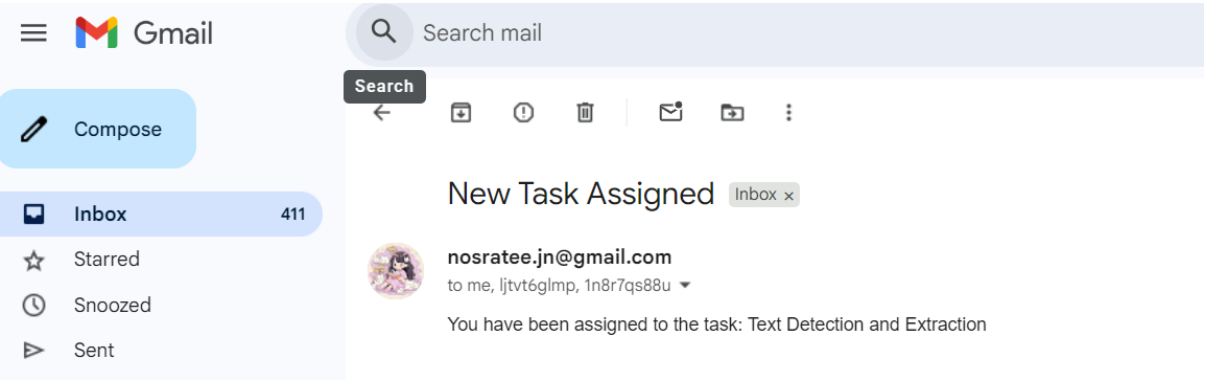
- **Task Details View:** Shows complete task information with quick status update.



- **Authentication & Authorization**
- **Admin Panel:** Django admin interface for managing users, tasks, and projects.



- **Email Notifications:** Automatically notifies employees when tasks are assigned.



- ## Sample Data Storage Overview in PostgreSQL

Project Table:

Data Output Messages Notifications

	id [PK] bigint	name character varying (200)	description text	start_date date
1	1	Cartooning an Image	nj	2026-01-02
2	2	Instagram Spam Detecti...		2026-01-31
3	3	Animal Species Prediction		2026-01-28
4	4	Teachable Machine		2026-02-28

Task Table:

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1

	id [PK] bigint	title character varying (250)	description text
1	5	Text Detection and Extraction	new
2	6	Stock Market Predictor	They are known for their accuracy based on mathematical assumptions and present circumstances
3	7	dgd	gd
4	8	Teachable Machine	Train a computer to recognize your own images, sounds, & poses.

TaskDetails:

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	id [PK] bigint	asset character varying (100)	priority character varying (1)	notes text	task_id bigint
1	1	tasks_asset/_Mastering_the_Art_of_Storytelling__Tips_for_Aspiring_Writers__.jpg	M		5
2	2	tasks_asset/default_img.jpg	H		6
3	3	tasks_asset/Open_old_book_on_table_with_abstract_business_illustration___Premium_Photo.jpg	L		7
4	4	tasks_asset/download-removebg-preview.png	H		8

User Table:

Data Output Messages Notifications

Showing rows: 1 to 7 Page No: 1 of 1

	username character varying (150)	first_name character varying (150)	last_name character varying (150)	email character varying (254)	is_staff boolean	is_active boolean	date_joined timestamp with time zone
1	nosratee	Naba		nosratee22@gmail.com	true	true	2025-12-27 16:57:44+06
2	naba	Nosratee Jahan	Naba	nosratee.jn@gmail.com	false	true	2026-01-02 18:29:07+06
3	Nishatee	Nishatee	B. Shahid	nosratee12@gmail.com	false	true	2026-01-03 00:19:46.085159+06
4	tourist	Tourist		ljtvt6glmp@mrotzis.com	false	true	2026-01-03 00:39:35+06
5	PetrMitrichev	Petr	Petr Mitrichev	1n8r7qs88u@cmhvzylmfc.com	false	true	2026-01-03 00:41:08+06
6	Errichto	Errichto		j7k7wsgtf8@mkzaso.com	false	true	2026-01-03 00:42:25+06
7	iahan	Nosratee Jahan	Naba		false	true	2026-01-10 10:00:53+06

Junction Table:

Data Output Messages Notifications

	id [PK] bigint	task_id bigint	user_id integer
1	5	6	11
2	6	6	12
3	7	6	13
4	8	5	11
5	9	5	12
6	10	5	13
7	11	7	11

3. Database Design

Major Entities

The Taskly system consists of the following four major entities:

1. **User (**Django's built-in User model.)
2. **Project**
3. **Task**
4. **TaskDetail**

Describe key attributes for each entity

Entity: User

- id: Unique system-generated identifier for each user (surrogate key)
- username: Unique login name
- email: Unique email address
- password: Encrypted user password
- first_name, last_name: Optional personal information
- is_active: Indicates whether the user account is active

Entity: Project

- id: Unique project identifier
- name: Name of the project
- description: Optional project details
- start_date: Project start date

Entity: Task

- id: Unique task identifier
- project_id: References the related project
- title: Short task title
- description: Detailed task description
- due_date: Task deadline
- status: Current task status
- created_at: Task creation time

- updated_at: Last modification time

Entity: TaskDetail

- id: Unique task detail identifier
- task_id: One-to-one reference to Task
- priority: Task priority (High, Medium, Low)
- notes: Optional additional notes
- asset: File or image attachment

Entity: Task_Assigned_To (Junction Table)

- id: Unique assignment identifier
- task_id: References Task
- user_id: References User

Primary Keys

Entity	Primary Key	Key Type
User	id	Surrogate Key
Project	id	Surrogate Key
Task	id	Surrogate Key
TaskDetail	id	Surrogate Key
Task_Assigned_To	id	Surrogate Key

All primary keys are surrogate keys (auto-generated IDs) created by Django ORM to ensure uniqueness, stability, and efficient relationship handling.

ER diagram

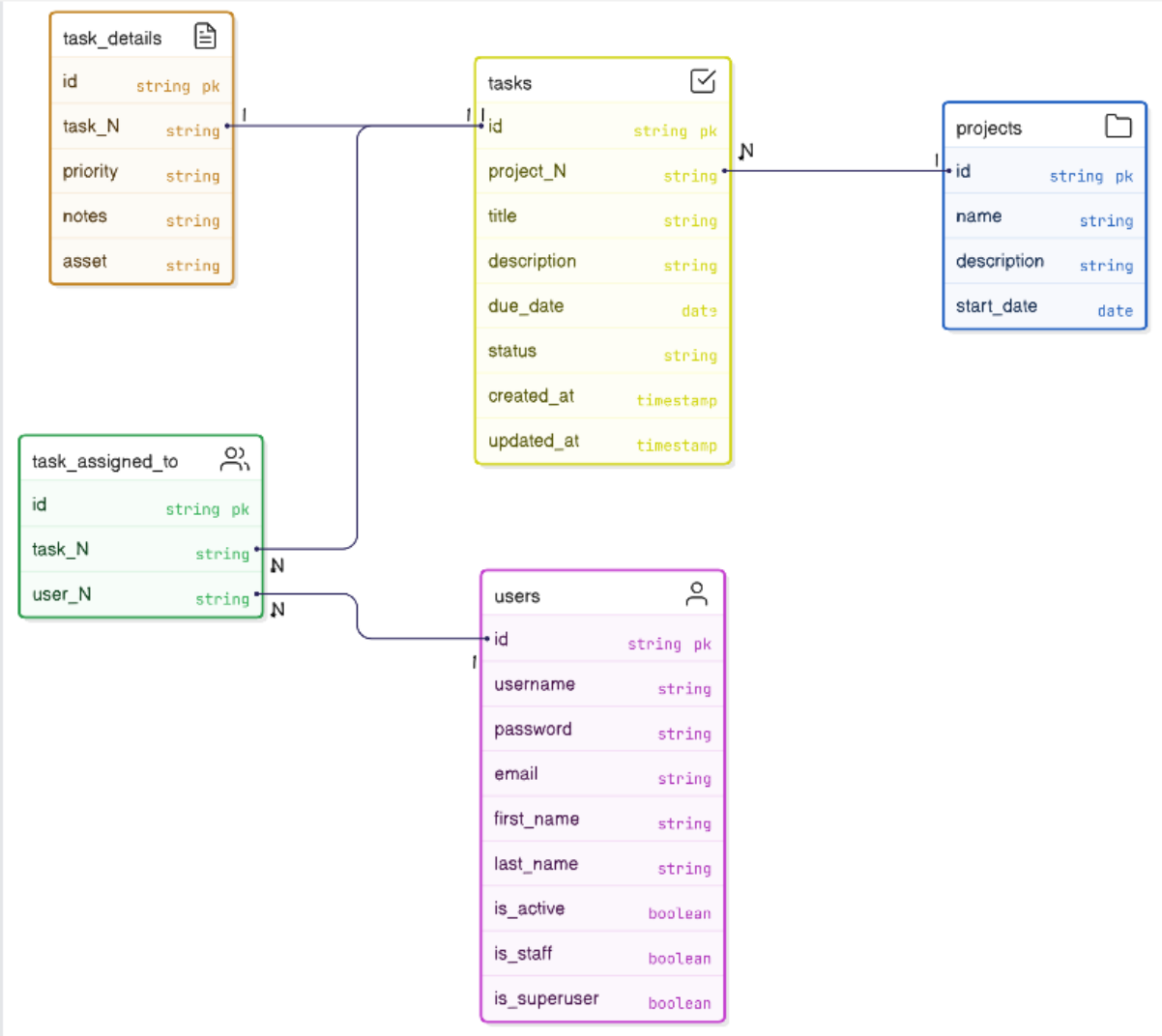


Figure : Entity Relationship (ER) Diagram

Relationship Descriptions

Relationship 1: Project → Task (One-to-Many)
One project can contain multiple tasks, while each task belongs to exactly one project.
This relationship is implemented using a foreign key (project_id) in the **Task** entity with **ON DELETE CASCADE**, ensuring that all tasks are removed when a project is deleted.

Relationship 2: Task → TaskDetail (One-to-One)
Each task has exactly one task detail record, and each task detail belongs to one task.
This relationship is enforced using a unique foreign key (task_id) in the **TaskDetail** entity with **ON DELETE CASCADE**.

Relationship 3: Task ↔ User (Many-to-Many)
A task can be assigned to multiple users, and a user can work on multiple tasks.
This relationship is implemented using a junction table (**Task_Assigned_To**) containing foreign keys to both **Task** and **User**, along with a composite unique constraint to prevent duplicate assignments.

Diagram Verification

The ER diagram accurately represents the system structure and constraints.

- All major entities are included: **User, Project, Task, TaskDetail, Task_Assigned_To**
- Primary keys and foreign keys are clearly marked
- Relationships and cardinalities (1:1, 1:N, M:N) are properly shown
- Constraints such as **NOT NULL, UNIQUE, and CASCADE** are correctly applied

4. Conclusion

Taskly successfully implements a complete task management system.

Core Functionalities Implemented:

1. Complete Task Management System

- Full CRUD operations (Create, Read, Update, Delete)
- Task creation with comprehensive details
- Multi-user task assignment
- Priority and status tracking
- File attachment support

3. Secure Authentication & Authorization

- Role-based access control (Manager, Employee, Admin)
- Django's built-in authentication system
- Group-based permissions

4. Advanced Features

- Automated email notifications using Django signals
- Real-time dashboard statistics.
- Query optimization (select_related, prefetch_related)
- Responsive design .
- Admin panel .

Limitations:

- Limited Notification System (Only email notifications)
- real-time collaboration
- No Task History/Audit Trail
- No Mobile App

Possible future improvements:

- Advanced Notifications System
- Task Dependencies & Gantt Charts
- Advanced Search & Filtering
- Multiple File Attachments
- Comprehensive Audit Trail
- Mobile Applications
- Comments & Discussions
- API Development